1.

$m_z = 2\,kg$

$p = 20\,bar$, $T = 590\,K$

$p_{0E} = 1\,bar$, $T_{0E} = 290\,K$

$m_t = 1\,t = 1000\,kg$

$h = ?$, $c_v = \frac{c_p}{\kappa} = 4{,}14\,kJ/kg\,K \Rightarrow R = 283$

$$w_{max} = u - u_{0E} - T_{0E}(s - s_{0E}) + p_{0E}(v - v_{0E})$$

$$= c_v(T - T_{0E}) - T_{0E}\left(c_p \ln\frac{T}{T_{0E}} - R\ln\frac{p}{p_{0E}}\right) + p_{0E}\left(\frac{RT}{p} - \frac{RT_{0E}}{p_{0E}}\right)$$

$w_{max} = 181{,}3\,kJ/kg$

$W_{max} = m \cdot w_{max} = 362{,}6\,kJ$

$W_{max} = m \cdot g \cdot h$

$$h = \frac{W_{max}}{m \cdot g}$$

$$\boxed{h = 36{,}96\,m}$$

② $P_t = 100\,MW$

$p_1 = 8\,MPa$, $T_1 =$ _____

$p_2 = 8\,MPa$, $T_2 = 453\,K$, $h_2 = 3350\ kJ/kg$

$p_3 = 0{,}7\,MPa$, $T_3 =$ _____

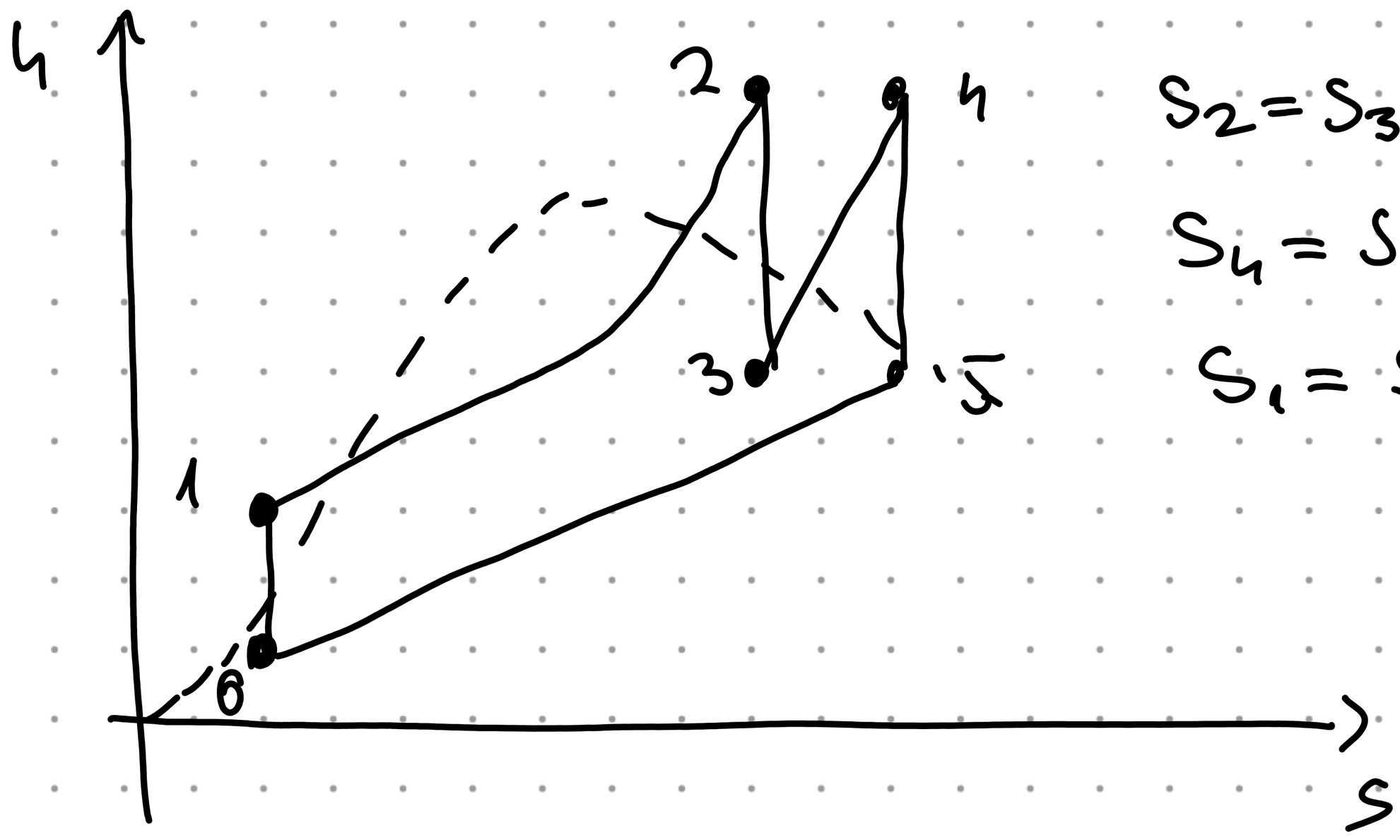
$p_4 = 0{,}7\,MPa$, $T_4 = 713\,K$, $h_4 = 3353\ kJ/kg$

$p_5 = 8\,kPa$, $T_5 =$ _____

$p_6 = 8\,kPa$, $T_6 =$ _____, $h_6 = h' = 174\ kJ/kg$

---

a) x_{VT}, b) x_{NT}, c) $\dot{m}$, d) $\eta_t = ?$



$s_2 = s_3$

$s_4 = s_5$

$s_1 = s_6$

a) $x_{VT} = \frac{s_3 - s'}{s'' - s'} = \frac{6,662 - 1,992}{6,705 - 1,992} = \underline{\underline{0,99}}$

b) $x_{NT} = \frac{s_5 - s'}{s'' - s'} = \frac{7,458 - 0,593}{8,230 - 0,593} = \underline{\underline{0,94}}$

c) $\dot{m} = ?$

$$w_T = w_{VT} + w_{NT} = h_2 - \overset{?}{h_3''} + h_4 - \overset{?}{h_5''}$$

$$h_3 = h' + x_{VT}(h'' - h') = 2741 \ kJ/kg$$

$$h_5 = h' + x_{NT}(h'' - h') = 2432 \ kJ/kg$$

$$w_T = 1530 \ kJ/kg$$

$$\dot{m} = \frac{P_T}{w_T} = \underline{\underline{65,36 \ kg/s}}$$

d) $\eta_t = ?$

$$w_p = v \cdot \Delta p = 0,001 \cdot (8 \cdot 10^6 - 8 \cdot 10^3)$$

$$w_p = 7,992 \ kJ/kg \approx 8 \ kJ/kg$$

$$h_1 = h_6 + w_p = 182 \ kJ/kg$$

$$\eta_t = \frac{w}{Q_{dod}} = \frac{w_{VT} + w_{NT} + w_p}{Q_{12} + Q_{34}} = \frac{h_2 - h_3 + h_4 - h_5 + h_6 - h_1}{h_2 - h_1 + h_4 - h_3}$$

$$\eta_t = 1 - \frac{h_5 - h_6}{h_2 - h_1 + h_4 - h_3} = \underline{\underline{0,4026}}$$

3. $P_j = 3990\ MW$

$m_{UO_2} = 105\ t$

$\Phi = 2{,}96 \cdot 10^{17}\ \frac{n}{m^2 s}$

$\sigma_f = 580 \cdot 10^{-28}\ m^2$

$H = 200\ MeV/fisija$

$c_H = 5{,}9\ kJ/kgK$

$T_{UL_H} = 568\ K$, $T_{IZ_H} = 600\ K$

$h_{UL_P} = 650\ kJ/kg$, $h_{IZ_P} = 2760\ kJ/kg$

$\dot{m}_P = 475\ kg/s$

---

a) $P_T = ?$, b) $P_P = ?$, c) $e = ?$, d) $\dot{m}_H = ?$

a) $P_T = h \cdot m_P \cdot \Delta h = \underline{4009\ MW}$

b) $P_P = \frac{P_T - P_j}{h} = \underline{4{,}75\ MW}$

c) $P_j = 200 \cdot 1{,}6 \cdot 10^{-13} \cdot N \cdot \sigma_f \cdot \Phi$

$\Rightarrow N = 7{,}263 \cdot 10^{27}$

$N = e \cdot m_{UO_2} \cdot \frac{238}{270} \cdot \frac{N_A}{235} \Rightarrow \underline{e = 0{,}03}$

d) $\dot{m}_H = ?$

$$\dot{m}_H \cdot C_H \cdot \Delta T = P_j$$

$$\dot{m}_H = \frac{P_j}{C_H \cdot \Delta T}$$

$$\underline{\dot{m}_H = 21\ 133\ kg/s}$$

4.

$$Q_{sr} = \frac{4}{5} \cdot (1200 - H)$$

$H_i = 750\,m$ u.v.

$H_u = 100\,m$ u.v.

---

a) $W_{bruto} = ?$

$$W_{bruto} = 8760 \cdot 9{,}81 \cdot \overset{1000\,kg/m^3}{\overset{\shortparallel}{\rho}} \cdot \int_{H_u}^{H_i} Q_{sr}\, dH$$

$$= 34{,}6\ TWh$$

b) $H_B = 80\,m$, $H_z = 400\,m$ $\Big\}$ $H_u = 80\,m$

$\eta = 0{,}9$

$P = ?$

[redacted]

$$P_{HE} = 9{,}81 \cdot \rho \cdot Q \cdot H_u \cdot \eta$$

$$= 9{,}81 \cdot 1000 \cdot \frac{4}{5}(1200 - 400) \cdot \text{[redacted]}80 \cdot 0{,}9$$

[redacted]

$$P_{HE} = 452\,MW$$

c) $H_z = 400\,m.\,n.v.$

$H_B = 80\,m$

$H_{DV} = 200\,m$

$\eta = 0{,}87$

$P_{HE} = ?$

$H_n = H_z + H_B - H_{DV} = 280\,m$

$P_{HE} = 9{,}81 \cdot 1000 \cdot \frac{h}{s}(1200-400) \cdot 280 \cdot 0{,}87$

$P_{HE} = 1{,}53\ GW$

[redacted]

d) $Q_{min} = 50\ m^3/s$

$P_{HE} = 9{,}81 \cdot 10^3 \left[\frac{h}{s}(1200-400) - 50\right] \cdot 280 \cdot 0{,}87$

$P_{HE} = 1{,}41\ GW$

---

e) $W_{god} = 8760 \cdot P \cdot 0{,}65$

$W_{god_c} = 8{,}7\ TWh$

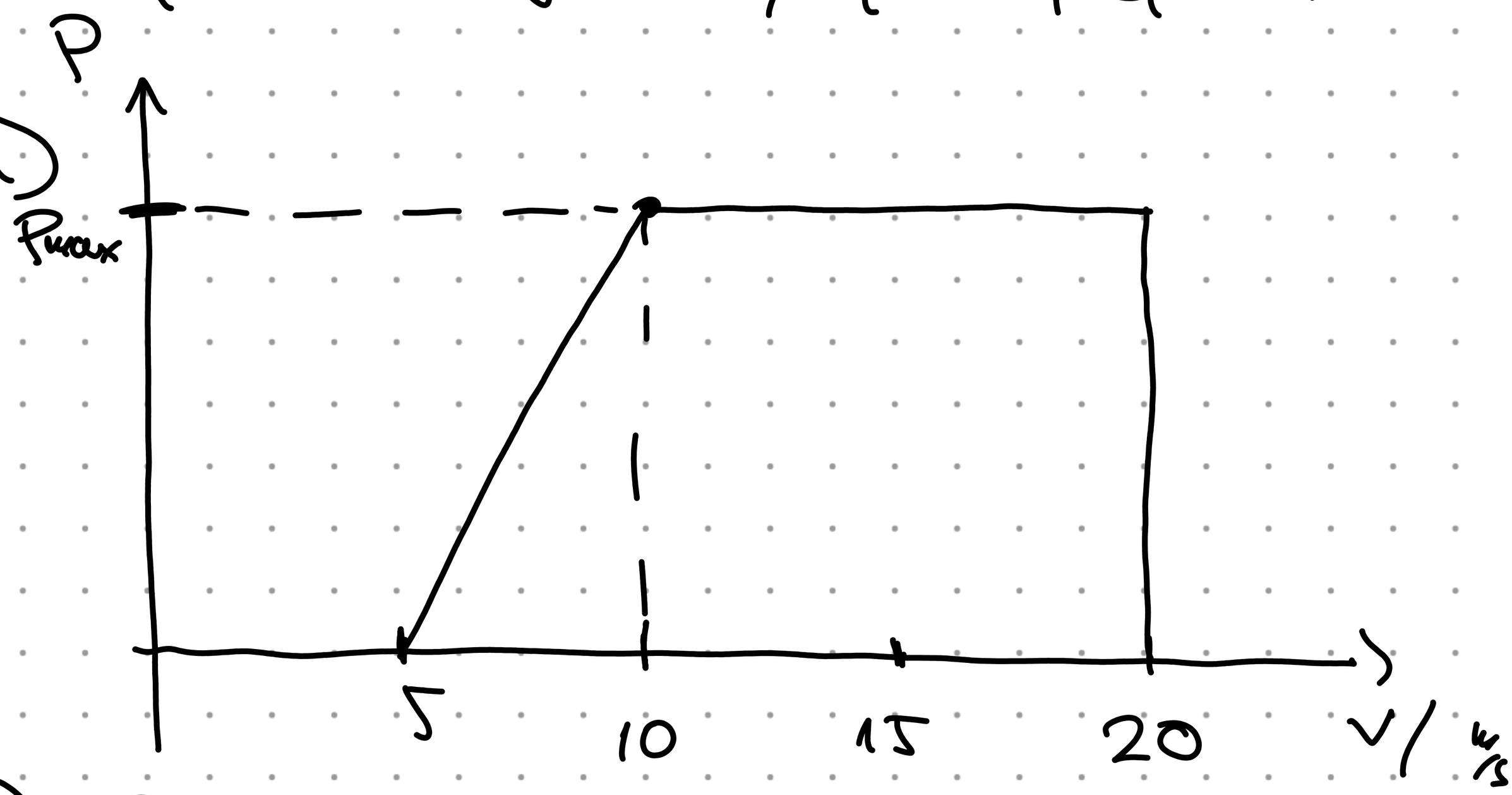
$W_{god_d} = 8{,}03\ TWh$

5. $r = 30\,m \Rightarrow A = 2827\ m^2$

$V_n = 10\,m/s$, $V_{min} = 5\,m/s$, $V_{max} = 20\,m/s$

$\rho = 1{,}225\,kg/m^3$, $\eta = 1$, $C_{pc} = 0{,}593$

a)



b) $P_5 = \eta \cdot C_p \cdot 0{,}5 \cdot \rho \cdot A \cdot v^3$

$= 1 \cdot 0{,}593 \cdot 0{,}5 \cdot 1{,}225 \cdot 2827 \cdot 5^3$

$= \underline{128{,}4\ kW}$

c) $P_{15} = \eta \cdot C_p \cdot 0{,}5 \cdot \rho \cdot A \cdot v_n^3$

$= 1 \cdot 0{,}593 \cdot 0{,}5 \cdot 1{,}225 \cdot 2827 \cdot 10^3$

$= \underline{1{,}03\ MW}$

c)

$$W_{god} = 8760 \cdot (0{,}2 \cdot P_5 + 0{,}15 \cdot P_{10} + 0{,}1 \cdot P_{15} + 0{,}2 \cdot P_{20})$$

$$W_{god} = 4\,285{,}2 \text{ MW}$$

$$(P_{10} = P_{15} = P_{20})$$

d) $u = ?$

$$u = \frac{W_{god}}{8760 \cdot P_n}$$

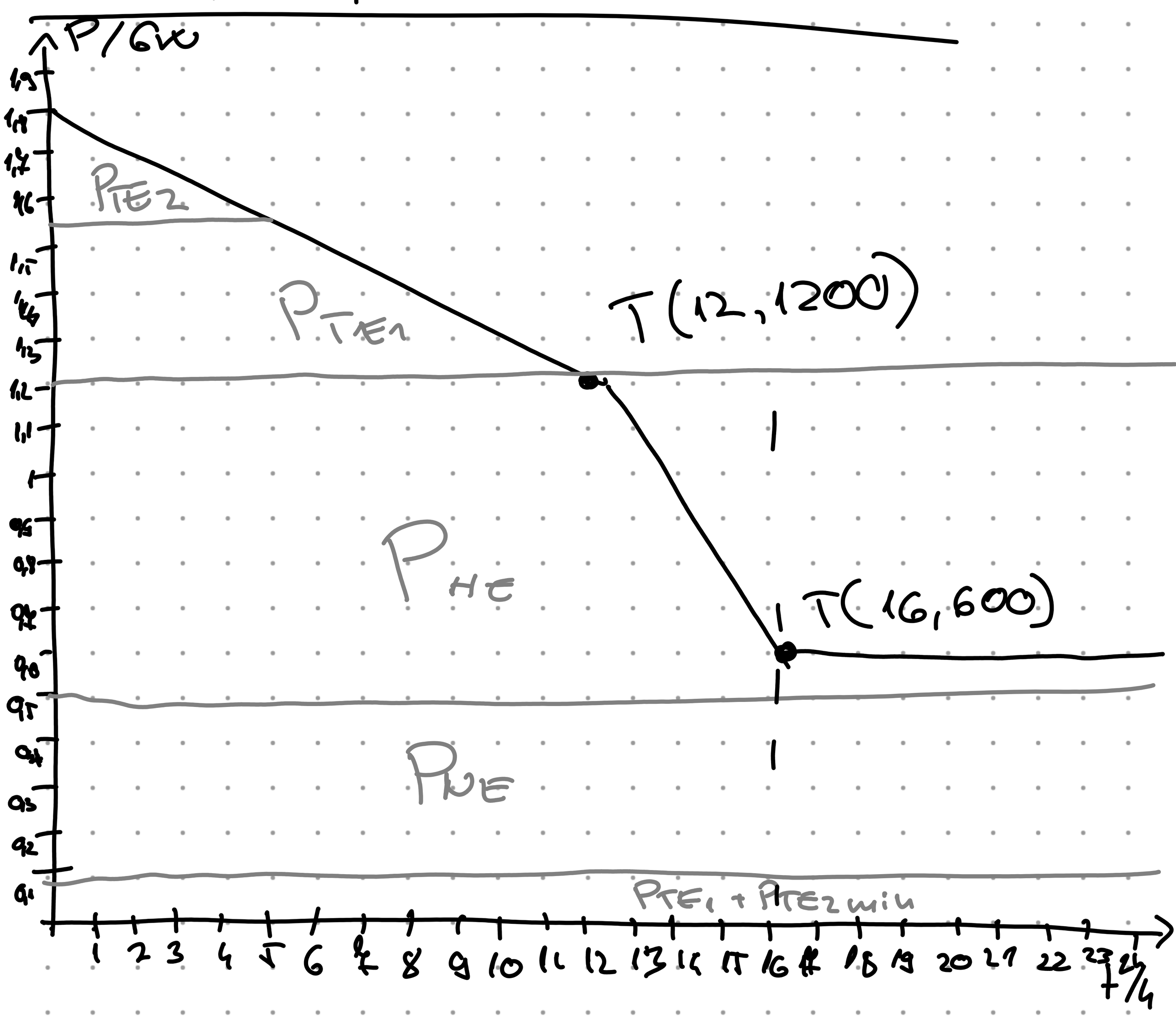
$$\underline{u = 0{,}475}$$

6. $P_{max} = 1500\,MW$ } $P_V = 1200\,MW$
$P_{min} = 600\,MW$

$T_{min} = 6\,h$

$W_V = 12\,000\,MWh$

$\beta = 0{,}5$



$$\alpha = 2 \cdot \frac{W_V}{T_V \cdot P_V} - \beta$$

$$= 2 \cdot \frac{12\ 000}{(24-8) \cdot 1200} - 0{,}5$$

$$= 0{,}75$$

T.P. $(x, y) = (\alpha \cdot T_V, P_V + \beta \cdot P_V)$

$$= (12, 1200)$$

b) $u = ?$

$$W_{VE} = W_V + W_{min} = 26400\ MW$$

$$u = \frac{W_{VE}}{24 \cdot P_{max}} = 0{,}611$$

c) $T_1(0, 1800)$, $T_2(12, 1200)$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$y - 1800 = \frac{1200 - 1800}{12 - 0}(x - 0)$$

$$P = 1800 - 50t$$

$$1750 = 1800 - 50t \Rightarrow t = 5h$$

$$W_{TE2} = 24 \cdot P_{TEmin} + \frac{5 \cdot 250}{2} = 1825\ MWh$$

d) $W_{preljev} = ?$

$$W_{preljev} = \frac{(16-12)(1200-600)}{2} + (24-16)(1200-600)$$

$$= 6000\,MWh$$

e) $T_{max} = ?$

$$T_{max} = \frac{W_{bz}}{P_{max}}$$

$$\boxed{T_{max} = 14{,}68\,h}$$